

Diffusion-based Kriging Model with Graph-enhanced Attention

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Abstract

In web-based systems, elements are commonly organized within a graph structure, with each node collecting essential spatio-temporal data. Examples include websites on the World Wide Web, traffic monitors in transportation networks, or sensors in the Internet of Things (IoT). However, sensors are typically deployed sparsely and unevenly, leaving the remaining nodes unobserved. The spatio-temporal kriging task, which infers values at unobserved nodes from observed ones, has thus attracted significant research interest. Due to limitations such as reliance on static graph structures and iterative Graph Convolution Network (GCN) frameworks, accurate kriging remains challenging. To address these issues, we propose a Diffusion-based Kriging Model with Graph-enhanced Attention (DKM-GA). Our approach first introduces a graph-enhanced attention mechanism that dynamically learns more accurate graph structures by combining predefined graph knowledge with global node value similarities. It is then integrated into a diffusion-based framework, which is tailored for the reliance of attention on known values. Therefore, the framework progressively refines the target values using correlated nodes, and the graph-enhanced attention selects more relevant neighbors based on the refined values. Furthermore, a node-based rescaling strategy is introduced to align the inference phase graphs to the training ones. Experiments on eight real-world datasets demonstrate that DKM-GA achieves superior performance, reducing estimation errors by up to 12.66%. Moreover, our analysis identifies three practical scenarios where the model delivers greater performance gains, even achieving 19.51% improvements on datasets that show minor gains under standard settings. These results highlight the effectiveness and potential of our model, while the scenarios provide settings for more comprehensive evaluations in terms of performance and robustness.

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CCS Concepts

• **Information systems** → **Spatial-temporal systems**; **Temporal data**; • **Mathematics of computing** → *Time series analysis*.

Keywords

Spatio-temporal, kriging, inductive learning, diffusion model

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1 Introduction

Sensors yield essential spatio-temporal data for a multitude of applications [18, 20, 41, 44], capturing the states of multiple dynamic systems [12]. For instance, in location-based social networks [12, 25, 42] web flow facilitates social network analysis, while data from traffic monitors [20, 44] provides critical insights into traffic conditions. However, due to challenges in sensor placement or web flow tracking [43], together with the high operational costs [19], typically these devices are sparsely and unevenly distributed. Consequently, available data is often inadequate for services that require fine-grained information at unobserved locations (represented as unobserved nodes in graphs). Therefore, Spatio-Temporal Kriging task [35] has been established to infer values at unobserved nodes from observed ones.

Spatio-temporal kriging methodologies are broadly categorized into two streams: inductive and transductive. Prevalent inductive approaches typically leverage GCNs [36], which are regarded as generalizable across different graph structures [1, 35]. Conversely, transductive models, such as tensor completion techniques [2, 8], are only applicable when the target unobserved nodes are known and fixed during training. This constraint makes transductive methods inflexible and recent studies [35, 38, 43] have demonstrated significant progress in spatio-temporal kriging particularly within inductive methods. However, several critical challenges persist, which are overlooked or only partially resolved by previous work.

Challenge 1: Predefined and static graph structures often fail to capture global and dynamic dependencies within data. Accurately identifying the most relevant neighboring nodes remains a major challenge. Beyond the graph matrices based on distance, auxiliary graphs for GCN typically requires careful, dataset-specific design and often depends on external information [43]. Although learned graph methods [15, 37] have shown promise in related tasks like forecasting, these methods generally assume accurate and complete input sequences, an assumption violated in kriging task with inherent missing values.

Challenge 2: Iterative GCN frameworks suffer from inherent limitations that can impair performance. Most current models rely on iterative GCN architectures when completing missing data from observed neighbors. However, as shown in Appendix B, iterative GCN frameworks inherently faces limitations, such as depth requirement and learnability issues: As data scales and missing rates increase, larger missing blocks necessitate deeper GCNs, which in turn require more training samples to sustain performance. This inefficiency highlights a fundamental weakness especially when unobserved nodes are unevenly distributed. It not only increases the need for training data and training time, but also aggravates oversmoothing issue, further reducing accuracy.

Challenge 3: The graph gap issue introduces multiple challenges, some of which remain challenging for current strategies. The *graph gap issue*, where the inference graph is denser than that used in training, can degrade performance. Solving this issue requires alignment between training and inference graphs. Current state-of-the-art, the *Incremental Training* strategy [38] augments the original graph with virtual nodes to match degree distributions. However, degree-based alignment overlooks edge weights, and the method's effectiveness relies on two manually designed factors: the rules for placing virtual nodes and the assumed missing rate determining their quantity. This heavy reliance on handcrafted parameters restricts the method's generalizability and effectiveness.

To tackle the aforementioned challenges, we equip our proposed DKM-GA model with three key innovations:

(1) We first select the most relevant neighbors dynamically to form the graph structures, instead of static graphs. It adopts the predefined graphs as prior knowledge, together with the attention based on estimated series values. By **adaptively learning how to incorporate a predefined graph into the attention matrices**, it directly addressing *Challenge 1*.

(2) We then propose a diffusion-based framework instead of the iterative GCN approach, which **offers several key advantages**. First, the diffusion process can be interpreted as an iterative Expectation-Maximization-like procedure: the E-step refines missing values from random initialization conditioned on correlated nodes, while the M-step utilizes the aforementioned attention mechanism to select more relevant neighbors based on the estimates. This design liberates the attention mechanism from its dependence on known values. Second, unlike general EM-style methods, diffusion models incorporate a structured noise scheduling mechanism. As value refinement from random initialization introduces varying noise levels during iterations, the diffusion steps provide essential noise-scale information to guide graph learning. Finally, the iterations of diffusion model enables information aggregation from distant nodes, and the noise scheduling prevents the prediction

from oversmoothing. Therefore each E-step employing shallow convolution layers, together with the one-step propagation in M-step, both address *Challenge 2* by efficient message passing without the strict depth requirements for GCN architectures.

(3) Moreover, we adopt a node-based rescaling strategy, which **aligns training and inference graphs based on graph weight distributions** instead of solely degrees. Combined with the inherent robustness of attention to structural shifts, it effectively mitigates the *graph gap issue* (*Challenge 3*).

Extensive experiments show that our model achieves state-of-the-art performance, outperforming most existing baselines. The principal contributions of this work are as follows:

- We propose a novel diffusion-based kriging model (DKM-GA). Our approach iteratively denoises target values conditioned on the related series, allowing the graph learning mechanism to identify related neighboring nodes. This mutual enhancement between value refinement and neighbor selection leads to more accurate predictions.
- We design graph-enhanced attention for dynamic graph learning, which adaptively integrates learned dependencies with predefined graph structures. By incorporating the noise-scale information from the diffusion step, this mechanism offers a more accurate modeling of spatial interactions. Carefully designed graph learning module and the node-based rescaling strategy ensure compatibility with the diffusion-based framework.
- Extensive experiments on 8 real-world datasets demonstrate that DKM-GA consistently outperforms existing methods, with improvements of up to 12.66%. In three additional practical scenarios, the model exhibits greater robustness, with performance gains as high as 19.51% on datasets that show minor improvements under standard settings. These results confirm the model's effectiveness under more comprehensive and realistic evaluation conditions, while proposing more thorough evaluation settings for future research.

2 Related Work

Spatio-Temporal Kriging is a widely used technique for spatial interpolation [11, 17], which involves predicting the value at an unobserved locations based on the observed values at nearby locations. Recent deep learning approaches have demonstrated significant advancements, achieving superior performance while relaxing traditional Gaussian assumptions [7]. Along them the inductive models show exceeding capabilities [35, 38, 43], with less dependence on the fixed unobserved locations in transductive setting [2, 5, 8]. For instance, IGNNK [35] constructs spatial graphs based on distance metrics by viewing the locations as nodes, and employs GCNs to recover values at unobserved locations. Subsequent models including LSJSTN [14] and INCREASE [43] have extended this paradigm by developing more informative graph structures and more expressive model architectures. KITS [38] identifies the challenge of *graph gap issue* where inference-phase graphs diverge from training-phase ones. While KITS introduces synthetic virtual nodes to address this issue, this solution imposes additional structural assumptions and creates reliance on presumed missing rates. Besides, current

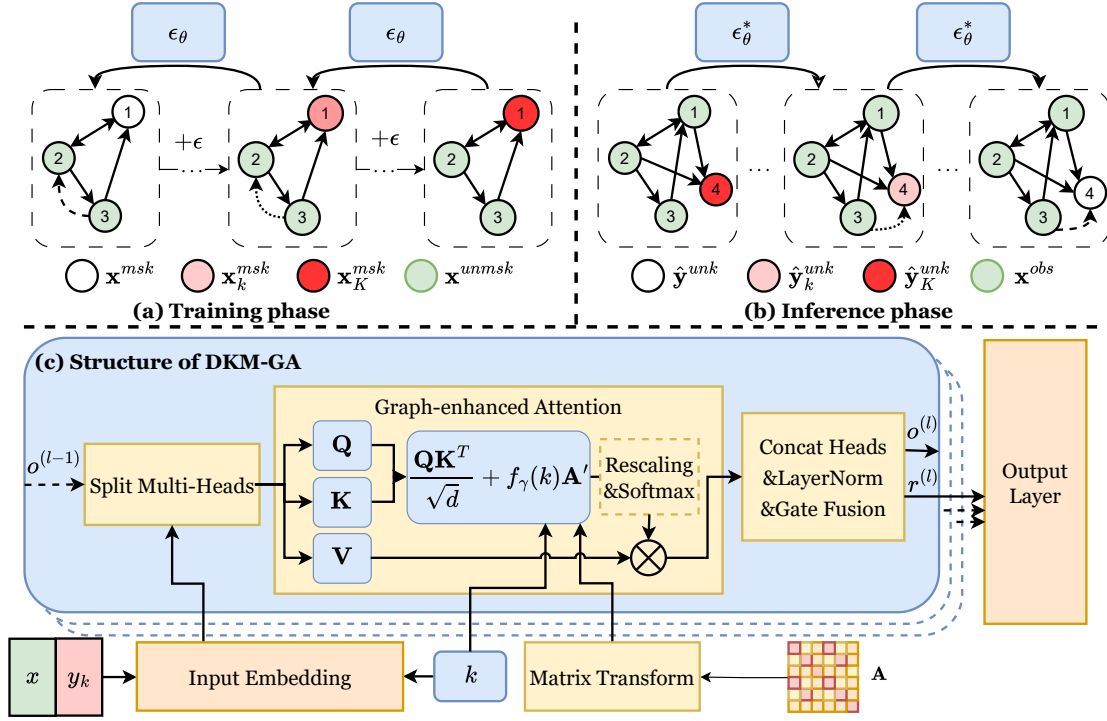


Figure 1: Overall architecture of DKM-GA. (a) and (b) are the illustration of the training and inference phase respectively. They depict the process how our model refines the values from the neighbors and adjusts the graph edges. (c) shows the structure of denoising model $\epsilon_\theta(y_k, x, G, k)$, with x as the input and y_k as the noisy target values. Notably, the ϵ_θ adopted in (a) is $\epsilon_\theta(x_k^{\text{msk}}, x^{\text{unmsk}}, G, k)$ and ϵ_θ^* in inference phase is the denoising model $\epsilon_\theta^*(y_k^{\text{unk}}, x^{\text{obs}}, G, k)$ with the graph weights rescaled.

methods exhibit limited robustness to structural shifts and focus exclusively on localized neighborhoods within predefined graphs. We demonstrate that a strategy effectively integrating dynamic and predefined graphs, not only mitigates *graph gap issue*, but also enhances kriging performance across diverse settings.

Diffusion Models for Time Series Modeling has been widely used in recent works. As powerful generative models, diffusion methods have been extensively applied across numerous domains [13, 27]. Recent time series tasks using diffusion include forecasting [10, 23, 26] and imputation [29]. For instance, TimeGrad [23] uses an autoregressive diffusion framework conditioned on historical data, while CSDI [29], TimeDiff [26], and MG-TSD [10] employ non-autoregressive generation for improved efficiency. Although some methods incorporate attention mechanisms [30] along feature dimensions (i.e. across spatial nodes) as effective feature extractor, most primarily focus on capturing temporal dependencies between past and future values with future values as the target, treating spatial attention mainly as an encoder for known values. Furthermore, without explicitly incorporating prior knowledge like node correlations, such models often *fail to properly distinguish unobserved nodes*. In contrast, our approach innovatively utilizes diffusion models as an EM-like framework for kriging. Our denoising model integrates global information and predefined graph knowledge to learn accurate dynamic graphs, and the framework effectively reconstruct values at unobserved locations from random initialization.

3 Preliminaries

3.1 Spatio-Temporal Kriging

Spatial-temporal time series comprise N^{all} numerical sequences observed at T distinct time points. Series are represented as $\mathbf{X} \in \mathbb{R}^{N^{\text{all}} \times T \times C}$, with $\mathbf{X}_{(i,t)} \in \mathbb{R}^C$ denotes the measured values of i -th location at time step t . Spatial relationships are encoded through a graph structure $G = (V, A)$, where V represents the set of graph nodes with cardinality N^{all} , and $A \in \mathbb{R}^{N^{\text{all}} \times N^{\text{all}}}$ denotes the adjacency matrix quantifying dependency strengths between locations.

The spatio-temporal kriging task estimates values at N^{unk} unobserved locations using simultaneously observed values from N^{obs} locations over a time window of length L . We define V^{obs} as the set of observed nodes and $V^{\text{unk}} = V \setminus V^{\text{obs}}$ as the set of unobserved nodes. The estimation target is thus $\mathbf{y}^{\text{unk}} \in \mathbb{R}^{N^{\text{unk}} \times L \times C}$, given information from observed nodes $\mathbf{x}^{\text{obs}} \in \mathbb{R}^{N^{\text{obs}} \times L \times C}$.

3.2 Diffusion Model

We follow standard notation to briefly introduce the Denoising Diffusion Probabilistic Model (DDPM) framework as an example. The diffusion process begins with a forward Markov chain that gradually adds noise to the input data $\mathbf{x}_0$ according to: $q(\mathbf{x}_k | \mathbf{x}_{k-1}) := \mathcal{N}(\mathbf{x}_k; \sqrt{1 - \beta_k} \mathbf{x}_{k-1}, \beta_k \mathbf{I})$, where $\beta_k \in (0, 1)$ is a monotonically increasing hyperparameter representing the noise schedule. The Gaussian noise formulation permits a one-step sampling procedure:

$\mathbf{x}_k \sim q(\mathbf{x}_k | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_k; \sqrt{\alpha_k} \mathbf{x}_0, (1 - \alpha_k) \mathbf{I})$, where $\alpha_k = \prod_{z=1}^k (1 - \beta_z)$. This forward process progressively corrupts the data until the final step $\mathbf{x}_K$ becomes approximately $\mathcal{N}(0, \mathbf{I})$. Then the reverse diffusion process learns to reconstruct data through: $\mathbf{x}_{k-1} \sim p_\theta(\mathbf{x}_{k-1} | \mathbf{x}_k) = \mathcal{N}(\mathbf{x}_{k-1}; \mu_\theta, \sigma_\theta \mathbf{I})$, where the mean μ_θ can be reparameterized as $\frac{1}{\alpha_k} (\mathbf{x}_k - \frac{\beta_k}{\sqrt{1-\alpha_k}} \epsilon_\theta(\mathbf{x}_k, k))$. Here, ϵ_θ represents a trainable denoising function. In DDPM, the variance of $p_\theta(\mathbf{x}_{k-1} | \mathbf{x}_k)$ depends on β_k as $\tilde{\beta}_k^{1/2}$ where $\tilde{\beta}_k = \frac{1-\alpha_{k-1}}{1-\alpha_k} \beta_k, k > 1$ and $\tilde{\beta}_1 = \beta_1$, though other diffusion models can employ different variance parameterizations. The model learns to reconstruct the input distribution through minimizing various training loss [27, 29] describing the discrepancy between $\hat{\mathbf{x}}_{k-1}$ and $\mathbf{x}_{k-1}$.

4 Methodology

4.1 Overview

Figure 1 illustrates the overall architecture of DKM-GA, in which we first introduce the diffusion-based framework and the training strategy. In this framework, the denoising model progressively refines prediction based on learned graphs, and attention utilizes the refined prediction to select more relevant neighbors. The dedicated denoising model in the framework consists of two key components: (1) Graph-enhanced Attention, which adaptively aggregates information from both locally correlated and globally similar nodes, and (2) Node-based Rescaling, which intuitively reweights the attention matrix in the Attention module to align the adjacency matrices in inference with the ones in training.

4.2 Diffusion-based Kriging Framework

Focusing on the reverse process described in Section 3.2, we develop a conditional model that learns the target distribution based on observed nodes, formalized as $p(\mathbf{y}_{k-1}^{\text{unk}} | \mathbf{y}_k^{\text{unk}}, \mathbf{x}^{\text{obs}}, G, k)$. Following the methodologies of DDPM [13] and Denoising Diffusion Implicit Models (DDIM) [27], we implement the model through a conditional denoising function $\epsilon_\theta(\mathbf{y}_{k-1}^{\text{unk}}, \mathbf{x}^{\text{obs}}, G, k)$, with the distribution parameterized as: $\mu_\theta = \frac{1}{\alpha_k} (\mathbf{y}_k^{\text{unk}} - \frac{\beta_k}{\sqrt{1-\alpha_k}} \epsilon_\theta(\mathbf{y}_{k-1}^{\text{unk}}, \mathbf{x}^{\text{obs}}, G, k))$. To achieve more deterministic and more efficient generation, we adopt DDIM strategy [27], with setting σ_θ as 0 and employing implicit sampling, as detailed in Appendix D.

In spatio-temporal kriging problems, the absence of historical information for unknown nodes presents training difficulties since test nodes are unavailable during training. To address this, we implement a **masked learning strategy** where observed nodes are randomly partitioned into "masked" ($\mathbf{x}^{\text{msk}}$) and "unmasked" ($\mathbf{x}^{\text{unmsk}}$). The diffusion model thus learns the distribution of $\mathbf{x}^{\text{msk}}$ conditioned on $\mathbf{x}^{\text{unmsk}}$, effectively simulating the inference process for estimating $\mathbf{y}^{\text{unk}}$ from $\mathbf{x}^{\text{obs}}$.

Therefore, the training can be formalized as the following optimization problem:

$$\min_{\theta} \mathcal{L}(\theta) = \min_{\theta} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}), k} \|\epsilon - \epsilon_\theta(\mathbf{x}_k^{\text{msk}}, \mathbf{x}^{\text{unmsk}}, G, k)\| \quad (1)$$

where $\mathbf{x}_k^{\text{msk}} = \sqrt{\alpha_k} \mathbf{x}_0 + (1 - \alpha_k) \epsilon$. The inference process of the framework can be formalized as:

$$\hat{\mathbf{y}}^{\text{unk}} \sim p_\theta(\hat{\mathbf{y}}^{\text{unk}} | \mathbf{y}_K^{\text{unk}}, \mathbf{x}^{\text{obs}}) = p(\mathbf{y}_K^{\text{unk}}) \prod_{k=1}^K p_\theta(\mathbf{y}_{k-1}^{\text{unk}} | \mathbf{y}_k^{\text{unk}}, \mathbf{x}^{\text{obs}}, G, k) \quad (2)$$

with $p_\theta(\mathbf{y}_{k-1}^{\text{unk}} | \mathbf{y}_k^{\text{unk}}, \mathbf{x}^{\text{obs}}, G, k) = \mathcal{N}(\mathbf{y}_{k-1}^{\text{unk}}; \frac{1}{\alpha_k} (\mathbf{y}_k^{\text{unk}} - \frac{\beta_k}{\sqrt{1-\alpha_k}} \epsilon_\theta^*(\mathbf{y}_k^{\text{unk}}, \mathbf{x}^{\text{obs}})), \mathbf{0})$, where ϵ_θ^* as the denoising model with the graph weights rescaled.

4.3 Graph-enhanced Attention

As dynamic graph learning is commonly implemented by attention mechanism, we incorporate it into the denoising model to select and aggregate information of related nodes. However, existing denoising models are not designed for kriging task. To make the model aware of the nodes' prior knowledge and the targets' noise level, the spatio-temporal kriging requires a specialized architecture.

To address this need, we design a graph-enhanced attention as the core component of our denoising model, incorporating residual connections and gated fusion mechanisms to stabilize training. The attention module captures spatial dependencies by computing pairwise similarities between current series representations, while the predefined graph structure emphasizes connections between closely related nodes. Formally, representing the input as $\mathbf{x} \in \mathbb{R}^{N \times T \times C}$ (which corresponds to $\mathbf{x}^{\text{unmsk}} \parallel \mathbf{x}_k^{\text{msk}}$ during training and $\mathbf{x}^{\text{obs}} \parallel \mathbf{y}_k^{\text{unk}}$ during inference), the multi-head graph-enhanced attention component can be formalized as follows:

$$\begin{aligned} \mathbf{E}_t &= f(\mathbf{x}_t, t, k) \in \mathbb{R}^{H \times N \times C} \\ \mathbf{A}' &= \text{transform}(\mathbf{A}), \mathbf{A} \text{ is the adjacency matrix of } G \\ \mathbf{Q}_t &= \mathbf{E}_t \mathbf{W}_Q, \mathbf{K}_t = \mathbf{E}_t \mathbf{W}_K, \mathbf{V}_t = \mathbf{E}_t \mathbf{W}_V \\ \text{SAttn}(\mathbf{x}_t, k) &= \text{softmax} \left(\frac{\mathbf{Q}_t \mathbf{K}_t^T}{\sqrt{d}} + \gamma \mathbf{A}' \right) \in \mathbb{R}^{H \times N \times N} \end{aligned} \quad (3)$$

The input function is defined as $f(\mathbf{x}_t, t, k) = g_{\text{inp}}(\mathbf{x}_t) + \text{DE}(k) + \text{TE}(t)$, where g_{inp} is a linear projection, DE denotes the diffusion-step embedding, and TE represents the temporal embedding. k indicates the diffusion step, which quantifies the level of the predicted noise. The function *transform* removes the nodes not in the input, converts the distance-based adjacency matrix into the multi-head bounded matrix representing node correlations, and aligns the edge weight with the scale of attention matrices, as detailed in Appendix E.4. The graph-enhanced self-attention mechanism (SAttn) jointly learns global correlations based on value similarity and semantic dependencies derived from the distance matrix. γ adaptively controls how to incorporate the predefined graph into the attention matrices by scaling, and since the diffusion step modulates both noise extent and prediction refinement, we parameterize γ as a function of the diffusion step embedding: $\gamma = f_\gamma(k) = g_\gamma(\text{DE}(k))$, where g_γ denotes a linear projection. This H -dimensional parameter is shared during the whole matrix. To sum up, the $\text{SAttn}(\mathbf{x}_t, k)$ module captures spatial dependencies, enabling the Graph-enhanced Attention layer to select related neighbors, so that the framework can aggregate the related nodes and refine the values accurately.

Additionally, a gated fusion mechanism is incorporated to enhance training stability:

$$\begin{aligned} \mathbf{h}_t &= \text{LayerNorm}(\text{SAttn}(\mathbf{x}_t, k)\mathbf{V}_t + \mathbf{x}_t) \\ \mathbf{r}_t, \mathbf{o}_t &= \sigma(g_{\text{gate}}(\mathbf{h}_t)) \cdot \text{Tanh}(g_{\text{filter}}(\mathbf{h}_t)) \end{aligned} \quad (4)$$

where $\mathbf{h}_t$ denotes the output of the attention layer. Here, g_{gate} and g_{filter} are linear transformations that map $\mathbf{h}_t$ to $\mathbb{R}^{2C}$, while $\mathbf{r}_t$ and $\mathbf{o}_t \in \mathbb{R}^C$ serve as residual and output of this layer.

4.4 Node-based Rescaling

The attention mechanism is generally considered more robust to graph structure shifts, as it integrates global information with pre-defined connectivity patterns. However, analogous to the "length extrapolation" problem in natural language processing (NLP) [22], attention mechanisms can experience performance degradation as the attention matrix grows larger, because even adding unrelated nodes will make the spatial aggregation more uniform. Since common NLP extrapolation strategies like positional encoding are not directly applicable to spatial nodes, we adopt a node-based rescaling strategy to mitigate the effects of more nodes in inference graphs.

While previous approaches like KITS [38] align graphs by preserving features such as average node degree, such methods often overlook edge weights. For graph-enhanced attention after the *softmax*, each node's connections can be interpreted as a discrete probability distribution. We propose to maintain distribution entropy through rescaling, formulated as $\text{SAttn}(\mathbf{x}_t, k) = \text{softmax}\left(\lambda \left(\frac{\mathbf{Q}_t \mathbf{K}_t^T}{\sqrt{d}} + \gamma \mathbf{A}'\right)\right)$ with a scaling factor $\lambda < 1$.

Although the study in NLP [4] empirically gives a rescaling strategy as $\lambda = \log N$, it lacks further analysis. Inspired by the analysis in [28], we present a proof in Appendix C demonstrating that scaling the attention matrix proportionally to $\log N$ is a simple version that approximately enhances distribution stability. With λ set to 1 during training for N^{obs} nodes, the rescaling coefficient becomes $\frac{\log N}{\log N^{\text{obs}}}$, where N represents the current number of nodes. As N increases, λ acts as a temperature parameter that sharpens the attention distribution, counteracting the uniform tendency.

5 Experiments

Experimental results demonstrate the superiority and robustness of our model in addressing the aforementioned challenges, while verifying the efficiency (Appendix D), explainability (Section 5.4), parameter robustness (Appendix F), etc.

Challenge 1: Model predicts more accurately by capturing dynamic dependencies across neighbor nodes, even those not identified by predefined graphs, as validated in Section 5.2 and 5.4.

Challenge 2: Diffusion-based framework empirically surpasses iterative GCN frameworks, evidenced in Section 5.2. Moreover, it avoids the inherent depth requirement, as evidenced in 5.3.1, where the "block missing" pattern requires larger receptive fields of GCNs.

Challenge 3: Our approach achieves outstanding performance in mitigating the *graph gap issue*, as shown in Section 5.2. Further, it exhibits reduced dependence on assumed missing patterns and rates, an advantage substantiated by experiments in Section 5.3.

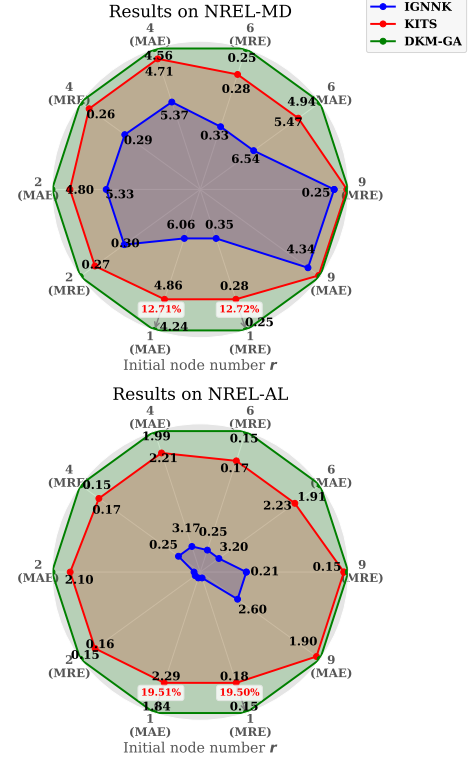


Figure 2: Comparison between our model and current state-of-the-art under block missing setting.

5.1 Experimental Settings

5.1.1 Datasets. Following the evaluation of [38], we conduct comprehensive experiments across eight diverse datasets. The datasets span multiple application domains: four on traffic systems (METR-LA, PEMS-BAY, SEA-LOOP, PEMS07), two on air quality monitoring networks (AQI-36, AQI), and two on solar power systems (NREL-MD, NREL-AL). Consistent with [43], we utilize only distance matrices due to data availability constraints. Detailed dataset descriptions are provided in Appendix E.1.

5.1.2 Baselines. We compare our proposed model against the following baseline methods: GLTL [2], MPGRU [5], and GRIN [5] as state-of-the-art transductive deep learning approaches; KCN [1], IGNNK [35], LSJSTN [14], INCREASE [43], and KITS [38] as leading inductive GCN-based methods. Although statistical baselines such as OKriging [7] are commonly used for inductive kriging, we exclude them here as they consistently demonstrate inferior performance compared to deep learning approaches. Comprehensive descriptions of all baseline methods are provided in Appendix E.2.

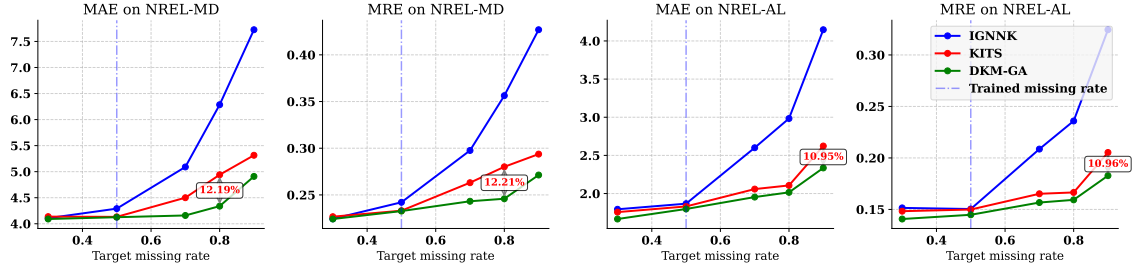
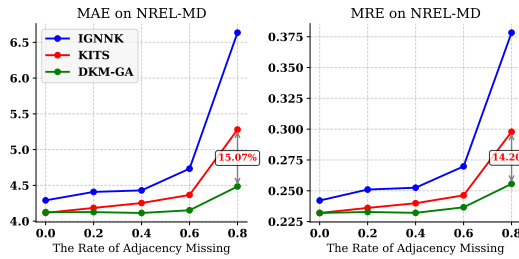
5.1.3 Evaluation Metrics. We employ three standard evaluation metrics for fair comparison: Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Mean Relative Error (MRE). Detailed formulations of metrics are provided in Appendix E.3.

5.1.4 Implementation Details. Our model is trained using the Adam optimizer with an initial learning rate of 0.002, mostly on an NVIDIA

Table 1: Comparisons with kriging baselines. "-" means some methods require the input of GPS coordinates or full distance information, which is not available in SEA-LOOP and PEMS07 datasets. The best results are shown in bold.

Model	AQI-36			AQI			NREL-AL			NREL-MD		
	MAE	MAPE	MRE	MAE	MAPE	MRE	MAE	MAPE	MRE	MAE	MAPE	MRE
GLTL(<i>NeurIPS'14</i>)	21.8970	0.6643	0.3073	30.9248	1.0133	0.4612	4.8230	1.5635	0.3885	12.4229	3.6326	0.7009
MPGRU(<i>ICLR'22</i>)	22.5312	0.9439	0.3126	22.7233	0.8289	0.3478	2.4439	1.8900	0.1968	5.3504	3.7190	0.3018
GRIN(<i>ICLR'22</i>)	16.7497	0.5235	0.2324	17.4716	0.5615	0.2674	1.9623	1.2376	0.1581	5.0114	2.5790	0.2827
KCN(AAAI'20)	20.6381	0.6190	0.2896	21.9771	0.6207	0.3278	4.6349	2.0685	0.3733	11.5863	4.5994	0.6537
IGNKK(AAAI'21)	22.3862	0.7892	0.3141	22.3997	0.7200	0.3341	2.1939	1.7267	0.1767	4.3950	2.4813	0.2480
LSJSTN(<i>ArXiv'21</i>)	22.7715	0.9022	0.3209	20.1396	0.5314	0.3003	2.0827	1.0960	0.1677	4.3206	1.9160	0.2436
INCREASE(WWW'23)	22.9031	1.0682	0.3214	19.9140	0.6130	0.2970	2.0936	1.1342	0.1686	4.7302	1.8028	0.2669
KITS(AAAI'25)	16.5892	0.3873	0.2302	16.2632	0.4187	0.2489	1.8315	0.7812	0.1475	4.1181	1.2523	0.2323
DKM-GA	14.6021	0.3564	0.2025	15.0265	0.3795	0.2300	1.7976	0.6970	0.1448	4.1255	1.2460	0.2327

Model	METR-LA			PEMS-BAY			SEA-LOOP			PEMS07		
	MAE	MAPE	MRE	MAE	MAPE	MRE	MAE	MAPE	MRE	MAE	MAPE	MRE
GLTL(<i>NeurIPS'14</i>)	8.6372	0.2627	0.1508	4.6986	0.1128	0.0741	-	-	-	-	-	-
MPGRU(<i>ICLR'22</i>)	6.9793	0.2223	0.1220	3.8799	0.0951	0.0620	4.6203	0.1393	0.0802	97.2821	1.1475	0.3177
GRIN(<i>ICLR'22</i>)	6.6096	0.1959	0.1155	3.8322	0.0845	0.0613	4.2466	0.1262	0.0743	95.9157	0.6844	0.3132
KCN(AAAI'20)	7.5972	0.2341	0.1326	5.7177	0.1404	0.0914	-	-	-	-	-	-
IGNKK(AAAI'21)	6.8571	0.2050	0.1197	3.7919	0.0852	0.0606	5.1865	0.1370	0.0901	80.7719	0.9314	0.2635
LSJSTN(<i>ArXiv'21</i>)	7.0666	0.2066	0.1234	3.8609	0.0836	0.0617	-	-	-	101.7706	0.8500	0.3325
INCREASE(WWW'23)	6.9096	0.1941	0.1206	3.8870	0.0835	0.0621	4.8537	0.1267	0.0842	93.7737	1.1683	0.3062
KITS(AAAI'25)	6.1276	0.1714	0.1071	3.5911	0.0819	0.0574	4.2313	0.1141	0.0734	75.1927	0.6847	0.2456
DKM-GA	6.1880	0.1741	0.1074	3.5376	0.0813	0.0565	3.8895	0.1030	0.0675	74.8504	0.5808	0.2444

**Figure 3: Comparison between our model and current state-of-the-art under inconsistent missing rate setting.****Figure 4: Comparison between our model and current state-of-the-art under incorrect adjacency matrix setting.**

GeForce RTX 3080 GPU. For main results, we set the missing rate to 50% across all datasets. Additional implementation details, including

hyperparameter configurations, adjacency *transform* choices, and training strategy, are described in Appendix E.4.

5.2 Main Results

We first compare the overall performance of our model against all baselines in Table 1, with the following observations:

(1) On most settings, our model achieves state-of-the-art performance, with improvements of up to 12.66% (in AQI-36). We attribute this superior performance to our Diffusion-based framework and Graph-enhanced Attention with rescaling strategy. The framework effectively enhances graph learning through refining the values to be more accurate. The attention mechanism captures both global and local information to form accurate graphs, enabling the fine-grained refinement of kriging results.

(2) Across the remaining settings, our model delivers competitive performance with minimal data dependence. The leading baseline, KITS, addresses the *graph gap issue* by inserting virtual nodes, but

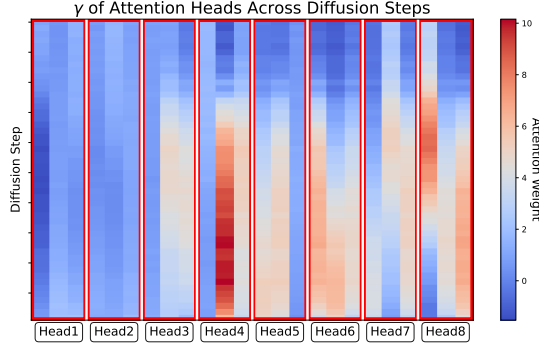


Figure 5: Visualization of γ weights for different diffusion steps, attention heads and layers.

depends on predefined rules and accurate missing rate estimation. Our solution, with its built-in rescaling strategy, provides superior robustness with less such constraints. The more limited performance gain can be explained by the benefit of KITS's virtual nodes and its additional nearest neighbor search, which are well-suited to these datasets.

(3) The improvement varies substantially across datasets, revealing our model's particular advantages in specific scenarios. For instance, on the AQI-36 dataset, our model outperforms all baselines by a significant margin. We attribute it to the exceptionally small node sets, where unobserved nodes are more likely to have fewer observed neighbors, a scenario where conventional methods struggle but our approach maintains strong performance.

In summary, our model demonstrates superior performance in kriging task while requiring fewer assumptions and predefined rules about input data. Furthermore, as indicated in observation (3), certain realistic scenarios exist where conventional baselines prove more inadequate. We therefore conduct additional experiments and analysis to further demonstrate our model's superiority.

5.3 Result Analysis on Further Settings

Analysis of the results, where our model shows exceeding advantages, reveals practical scenarios that may have been overlooked in previous studies. To demonstrate how these scenarios affect model performance, we conduct additional comparisons using NREL-AL and NREL-MD datasets, where our model and current state-of-the-art methods show comparable performance in Table 1. These further experiments help explain the superior performance and robustness of our model, and guide future investigations and evaluation under different settings.

5.3.1 Analysis of Block Missing Patterns. As previously discussed, some models get inferior results when unobserved nodes form connected clusters with limited neighbor information available. We therefore propose a scenario "Block Missing", where connected nodes tend to be missing simultaneously¹. This scenario represents a special case of "Spatially Clustered Missingness", in contrast to the commonly assumed "Missing Completely At Random (MCAR)"

¹Following the terminology of [38], we use the terms "missing" and "unobserved" interchangeably.

pattern [5, 24]. In real-world applications such as traffic sensor deployment or web user tracking, clusters of connected nodes may become unavailable simultaneously due to severe road environment or web route protection, validating its practical relevance. "Block Missing" is implemented by randomly selecting r initial nodes as missing, then iteratively selecting neighbors of unobserved nodes until reaching the target missing rate. Figure 2 presents kriging results on NREL-AL and NREL-MD², which demonstrate that while larger r values make the "Block Missing" scenario approach MCAR, our model achieves significant performance gains across various r and datasets. This superiority arises because traditional methods require D -layer convolutions to propagate information across a missing block of diameter $2D$, whereas our model directly incorporates global information from semantically similar nodes. Our model thereby overcomes *Challenge 2* and avoids the negative consequences of depth requirement such as the need of more training samples and the oversmoothing issue.

5.3.2 Analysis of Inconsistent Missing Rates. Several baseline methods heavily depend on predefined missing rates during training. However, in practical applications, true missing rate may be inconsistent with the assumed one, as the presumed rate can be incorrect and unobserved nodes can be incrementally added. We investigate this factor through the following experiment: models are trained using the default missing rate from previous experiments, while inference is conducted with varying missing rates. As shown in Figure 3, our model performs comparably to state-of-the-art methods when training and inference missing rates match. However, as the rate diverges, baseline performance deteriorates significantly while our model remains stable. This robustness stems from our framework and dynamic graph learning, which are robust to the inconsistency. The experiment results (along with the results in Section 2) validate that our model resolves *graph gap issue* by attention with node-based rescaling, and the mechanism is robust to different missing patterns and missing rates (*Challenge 3*).

5.3.3 Analysis of Incorrect Adjacency Matrices. This investigation was motivated by our model's exceptional performance on the SEA-LOOP dataset, which features a more sparse graph structure. We hypothesized that our method might demonstrate particular robustness to imperfect adjacency matrices. To quantitatively evaluate this, we systematically remove connections from the given adjacency matrix at varying rates and use these degraded matrices for spatial dependency learning. As illustrated in Figure 4, our model maintains strong and stable performance across various corruption rates, achieving improvements of up to 15.07% compared to baselines. This robustness can be attributed to our model's ability to aggregate global information, making it less dependent on the accurate adjacency matrix.

5.4 Ablation Study

We conduct comprehensive ablation studies to evaluate the contribution of each component in our model, with results summarized in Table 2. In our experimental design, the "Framework" is divided into two variants: an "Iterative-GCN" approach that directly predicts

²Since INCREASE constructs multiple adjacency matrices using additional external information, we compare our model specifically with KITS and IGNNK.

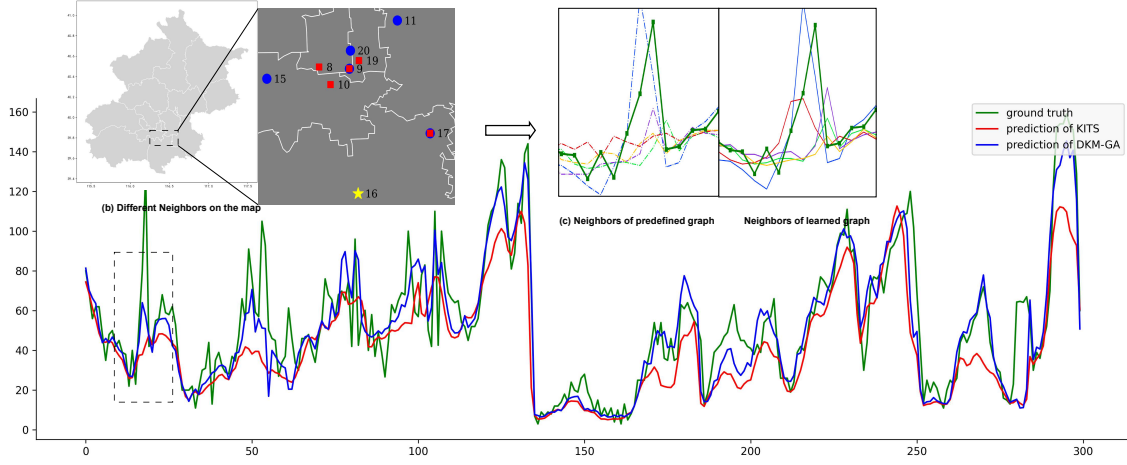


Figure 6: Visualization of kriging results. Figure(b) shows different nearest neighbors of the target (yellow node) in the dashed box, captured by predefined graph (red nodes) and learned graph (blue nodes). Figure(c) illustrates series values of different nodes, where the bold green one shows the target node and the ones with other colors show different neighbor nodes.

outcomes from iterative GCN, and a "Diffusion-based" approach that iteratively refines by the diffusion-based frameworks. We further examine the components of the Graph-enhanced Attention mechanism: (1) When the "Attention" component is ablated, we substitute it with a graph convolution layer while simultaneously removing other components (Exps.A and C). (2) When "Step-based γ " is disabled, we implement γ as a trainable parameter independent of other variables (Exps.D). (3) When the "Rescaling" mechanism is omitted, we employ the original attention matrix during inference (Exp.E). The comparison between Exps.A and C demonstrates substantial improvement attributable to the diffusion-based framework, showing the potential of diffusion models as an Expectation-Maximization process in kriging task. Subsequent comparisons among Exps.C, D, E, and F confirm the individual contributions of each module within the Graph-enhanced Attention mechanism.

Table 2: Performance of ablation study.

Exps.	Framework	Graph-enhanced Attention			AQI-36		NREL-MD	
		Attention	Step-based γ	Rescaling	MAE	MRE	MAE	MRE
A	Iterative GCN	✓	✗	✗	22.6265	0.3137	6.0967	0.3440
B	Iterative GCN	✓	✗	✓	19.1244	0.2652	4.1994	0.2369
C	Diffusion-based	✗	✗	✗	15.4773	0.2146	4.2060	0.2373
D	Diffusion-based	✓	✗	✓	14.7061	0.2039	4.1942	0.2366
E	Diffusion-based	✓	✓	✗	14.6170	0.2027	4.1402	0.2336
F	Diffusion-based	✓	✓	✓	14.6021	0.2025	4.1255	0.2327

To further validate the effectiveness of step-based γ , we visualize γ values across different diffusion steps and attention heads, as shown in Figure 5. Several key observations emerge from this visualization: (1) Different attention heads capture different correlation learning patterns. For instance, heads 1 and 2 demonstrate consistent γ values primarily capturing pairwise node similarities, while head 4 shows gradually increasing γ values that strengthen predefined graph connections. (2) Contrary to intuitive expectations, γ evolution follows non-monotonic patterns. Rather than prioritizing graph structure early in the denoising process (at larger diffusion

steps) before shifting to global similarities, the model initially leverages global information, while certain attention heads subsequently focus on graph structure. (3) The step-based γ mechanism effectively captures denoising dynamics and enables multi-head attention to represent diverse aspects of data correlation, demonstrating greater expressiveness than fixed weights or single trainable parameters and validating our design approach.

Furthermore, we analyze prediction results as illustrated in Figure 6. Our model demonstrates accurate performance, particularly in capturing series peak patterns. Figure (b) compares the predefined graphs with the learned patterns, revealing distinct connectivity structures. Neighbor values in Figure (c) show **our model selects neighbors exhibiting stronger peak pattern similarity**, with 3 of 5 related nodes overlooked by predefined graphs. Conversely, neighbors in the predefined graph but not preserved by our attention mechanism consistently demonstrate weaker pattern similarity. The visualization confirms the effectiveness of our approach in overcoming *Challenge 1* by capturing learned dynamic graphs.

6 Conclusion

We propose a diffusion-based model with graph-enhanced attention (DKM-GA), learning dynamic graphs for inductive spatio-temporal kriging. Extensive experiments on eight real-world datasets show that DKM-GA outperforms current state-of-the-art methods by up to 12.66%. We also establish three comprehensive and practical evaluation scenarios that demonstrate its superior robustness, with significant performance gains of up to 19.51%. Future work may focus on computational optimization for faster inference, and extending our approach to other data modalities, such as spatio-temporal interpolation for feature propagation in cell networks.

Acknowledgments

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A Relevance to Web

Spatio-temporal graphs have been extensively utilized in various web applications, such as social networks [25, 42] and web data mining [12]. However, in real-world scenarios, inferring values at unobserved nodes from observed ones on the web remains highly challenging, significantly limiting the applicability of existing methods in web and social graph contexts. We aim to address this issue by enhancing the performance of spatio-temporal kriging, therefore leverage the coarse-grained spatio-temporal web data as essential resource for social and web applications.

Furthermore, our model demonstrates superior performance under practical conditions often encountered in web environments, such as the "Block Missing" pattern—commonly arising due to route protection mechanisms applied to clusters of highly correlated web-sites, especially in Local Area Networks. It also handles imperfect data scenarios effectively, including "Inconsistent Missing Rate" and "Incorrect Adjacency Matrices" settings. We argue that the robustness of our model in these realistic settings not only enhances its practical applicability but also provides a more comprehensive evaluation framework for web-oriented kriging methods.

B Analysis of Iterative GCN Framework

In scenarios with data missing, the learnability of iterative Graph Neural Networks (GNNs, mostly GCN) framework can be limited [9]. It is because firstly, in iterative GNN with connection A , the iterative number of GNN (i.e. depth of GNN) $Depth$ must satisfy $Depth \geq \max_{z' \in V_{unk}} \min_{z \in V_{obs}} |SP(z, z')|$, with SP as the shortest path in G .

Following proposition 4.3 in [9], needed samples of data for desired Generalization Risk should satisfy $m \geq \frac{C_1}{C_2 + C_3 \cdot C_4^{Depth} \cdot Depth \cdot \rho^\tau}$, with $C_{1/2/3/4}$ as constants, ρ as the missing rate and τ as the maximum size of $Depth$ -hop neighborhood of any node. It states the iterative GNN framework may face the issue that larger ρ can cause smaller τ and larger $Depth$, leading to the need of more samples, deeper GNN framework, and probably more severe issue of over-smoothing, particularly on specific missing pattern. Conversely, our model enables one-step propagation for expressive distribution modeling, with better results.

C Analysis of λ Choice

We formalize the rescaled graph-enhanced attention $\lambda \geq 0$ as $SAttn(\mathbf{x}_t, k) = softmax\left(\lambda \left(\frac{Q_t K_t^T}{\sqrt{d}} + \gamma A'\right)\right)$, where the scaling factor λ depends on N . We choose λ proportional to $\log(n)$ to enhance distribution stability, based on the following analysis.

Analysis Denoting s_i as the graph weight of i -th node before $softmax$, i.e. $(\frac{Q_t K_t^T}{\sqrt{d}} + \gamma A')_i$, then $p_i(j) = \frac{\exp(\lambda s_i(j))}{\sum \exp(\lambda s_i(z))} \in \Delta^{N-1}$ corresponds to a discrete distribution with p_i as the value after $softmax$. The entropy of p_i , denoted as $Entropy(p_i)$ can be calculated as:

$$\begin{aligned} H(p_i) &= - \sum_j p_i(j) \log p_i(j) \\ &= \log \sum_{j=1}^N \exp(\lambda s_i(j)) - \lambda \sum_{j=1}^N p_i(j) s_i(j) \end{aligned} \quad (5)$$

Given

$$\begin{aligned} \log \sum_{j=1}^N \exp(\lambda s_i(j)) &\in [\lambda \max(s_i), \lambda \max(s_i) + \log N] \\ \log \sum_{j=1}^N \exp(\lambda s_i(j)) - \log N &\in [\lambda \text{mean}(s_i), \lambda \max(s_i)] \quad (6) \\ \text{and } \lambda \sum_{j=1}^N p_i(j) s_i(j) &\in [\lambda \text{mean}(s_i), \lambda \max(s_i)], \end{aligned}$$

therefore,

$$Entropy(p_i) \in [\max(0, \log N - \lambda(\max(s_i) - \text{mean}(s_i))), \log N] \quad (7)$$

From Equation 7 we can draw the following observations:

- With a fixed λ , generally the scale of $Entropy(p_i)$ increases with the increment of N , verifying our analysis that the increment of N may cause the output of $softmax$ tends to be more uniform. The upper-bound of $Entropy(p_i)$ is $\log N$, which can always be reached with any specific λ . As it indicates that the model can not distinguish all the nodes, we identify it as a rare case where almost all graph-based kriging models struggle.
- The lower-bound of $Entropy(p_i)$ also depends on $\log N$, which can be formalized as $\max(0, \log N - \lambda(\max(s_i) - \text{mean}(s_i)))$, which shows the different choice of λ may affect the distribution. When λ is smaller than the optimal one, $Entropy(p_i)$ is inherently larger than $\log N - \lambda(\max(s_i) - \text{mean}(s_i))$. With N in inference phase growing, it may be larger than the upper-bound in training phase, with significant distribution shift.
- As λ changes the scale of p_i , practically we should select the smallest λ to avoid exceeding scaling. Therefore, $\lambda^* \propto \frac{\log N}{\max(s_i) - \text{mean}(s_i)}$, when s_i is not identical. The analytical solution of the Entropy can not be directly calculated due to the complex $\max(s_i)$. However, in deep learning models, the approximate solution is acceptable, assuming s_i is sampled from the same distribution $\mathcal{N}(s; \mu, \sigma)$:

- (1) With finite and small N , $\max(s_i) - \text{mean}(s_i)$ tends to be stable, as intuitively most values from Gaussian distribution fall in range $[\mu - 3\sigma, \mu + 3\sigma]$, indicating approximately $\max(s_i) - \text{mean}(s_i) \sim 3\sigma$.
- (2) When $N \rightarrow \infty$, following Fisher-Tippett-Gnedenko theorem [6], the maximum satisfies

$$\lim_{N \rightarrow \infty} \frac{\mathbb{E}[\max(s_i) - \text{mean}(s_i)]}{\sqrt{2\sigma^2 \log N}} = 1 \quad (8)$$

indicating $\max(s_i) - \text{mean}(s_i) \sim \sigma \sqrt{2 \log N}$ with $N \rightarrow \infty$, with a refining factor $\text{refine}(N) = \frac{\log \log N + 2(\log \pi + 0.57721)}{2\sqrt{2 \log N}}$.

- (3) Therefore, a reasonable magnitude of λ can be selected from $\frac{\log N}{3\sigma}$ and $\sqrt{\frac{\log N}{2\sigma^2}}$. The former is easier to compute with empirically acceptable precision for $N \leq 3000$, with the latter more precise with more calculation on refining factor. As most time N is actually far from infinity, we follow the empirical choice in [4, 28] selecting λ as $\frac{\log N}{3\sigma}$. As in Attention layer, to make training more stable, λ in training phase is set to 1 to make the initial variance of s_i to be a constant. Rescaling factor λ dependent on N are thus set to be $\frac{\log N}{\log N^{\text{obs}}}$ as N^{obs} is the node number in training phase.

Thus, we assign $\lambda \propto \log N$ to make its distribution more stable. It should be emphasized that when the nodes are generally similar, and the distribution of s_i is similar to the uniform distribution, rescaling by λ may not improve the result, as the upper-bound of

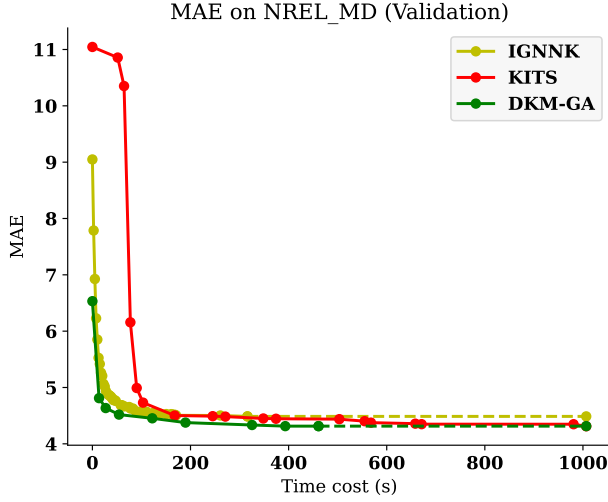


Figure 7: Visualization of the validation loss during the training process.

the Entropy can always be reached regardless of the choice of λ , which may explain the reason why sometimes rescaling strategy leads to less improvement.

D Model Complexity

The key component of model complexity in our model is the graph-enhanced attention module, whose computation complexity is $O(T \times N^2 \times d)$, which is the general complexity of kriging model. The detailed comparison is in Table 3, with K as the iteration of model calling, as the multiplication of sampling steps and sampling numbers.

DDIM (Denoising Diffusion Implicit Models) [27] is a novel sampling algorithm designed for diffusion models. Its acceleration technique lies in redefining the forward process as a non-Markovian chain, thereby freeing it from the constraint of a fixed, slow reverse process. It significantly accelerates the generation process by reconstructing the original denoising sampling—which required hundreds or thousands of iterative steps—into a deterministic process with less sampling steps. By introducing an adjustable variance parameter, DDIM allows the model to use a subsequence far shorter than the training steps (e.g., 10 steps) for sampling calculations without needing to retrain the model. This enables a speedup by drastically reducing the number of sampling iterations with almost no loss in output quality, as it effectively finds a "shortcut" or "fast path" from random noise to the generated image.

By adopting the strategy of DDIM, we can accelerate our model by setting sampling step to be 10, and by adopting σ_θ as 0 to make model deterministic, we simply set the sampling numbers as 1. As our model shows faster training speed, and more robust to overfitting, we propose to conduct the validation after 5 epochs of training, to get better efficiency in practice, as the strategy we utilized in our experiments. It is notable that the less validation does not significantly influence the results, as the results in our model is implemented with $K = 10$, and the validation interval setting

(denoted as $VI = 5$) ensures the similar magnitude of training time with KITS as illustrated in Table 3. Moreover, as our model tends to get converged to the optimal result in less epochs, with the rapid advancing of fast diffusion model, we believe our model as an efficient and promising method for kriging task. As illustrated in Figure 7, the validation loss of our model drops most quickly, converges earlier than all baselines to get the best result. In summary, in offline batch kriging or block-missing scenarios, DKM-GA's robustness and accuracy likely justify its cost. In latency-critical settings with moderate missing rates and reliable predefined graphs, a faster baseline like KITS/IGNNK may be the choice despite accuracy loss

Table 3: Time complexity of model and time consumption (10^{-2} s/epoch) on NREL-MD Dataset.

Model	Nodes	Training Complexity	Time	Nodes	Inference Complexity	Time	Avg. Time
IGNNK	N^{obs}	$O((N^{\text{obs}})^2 \times T \times d)$	0.97	N^{all}	$O((N^{\text{all}})^2 \times T \times d)$	0.73	250.8
INCREASE	N^{all}	$O((N^{\text{all}})^2 \times T \times d)$	5.88	N^{all}	$O((N^{\text{all}})^2 \times T \times d)$	2.67	1290.6
DKM-GA	N^{obs}	$O((N^{\text{obs}})^2 \times T \times d)$	4.20	N^{all}	$O(K \times (N^{\text{all}})^2 \times T \times d)$	26.1	1355.8

E Experimental Settings

E.1 Datasets

The statistics of all eight datasets are presented in Table 4. We follow the previous data pre-processing strategies, (i.e. 7/1/2 ratios to split the datasets except AQI and AQI-36, and take data from March, June, September and December as test set) in [38] for fair comparison. Besides the random missing setting, where model uniformly selects a specific rate of nodes as unobserved, we further propose a practical setting as "Block Missing", where several clusters of connected nodes are missing. We implement this setting by randomly selecting r initial nodes as unobserved nodes, then iteratively selecting neighbors of unobserved nodes until reaching the target missing rate. To make fair comparison, the target missing rate in "Block Missing" is the same as in Main Results as 50%. Moreover, the data normalization choice is selected according to the scale of the datasets. In PEMS07, NREL-MD and NREL-AL, we follow IGNNK [35] and KITS [38] to use min-max normalization, while the others use standard zero-mean normalization.

E.2 Baselines

We compare our model with the following baseline methods:

GLTL. The Greedy low-rank tensor learning (GLTL) [2] is a transductive tensor factorization for spatio-temporal kriging. The missing data in the input data (with tensor shape as $locations \times timesteps \times variables$) are recovered by tensor completion.

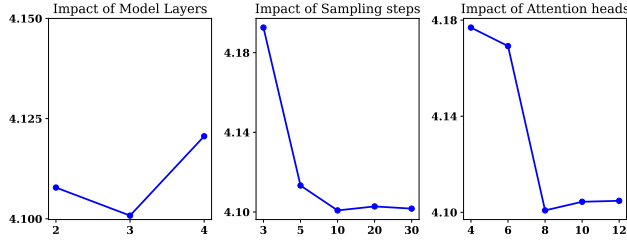
MPGRU and GRIN. GRIN [5] is a method target at time series imputation task incorporating GNN-based framework with Gated Recurrent Unit (GRU), which also covers the kriging task. MPGRU is a reduced version of GRIN, with different decoding layers.

KCN. Kriging Convolutional Network (KCN) [1] combines the K -nearest neighbors with GNN to interpolate one by aggregating the information of its neighbors.

IGNNK. IGNNK [35] takes the spatio-temporal information of nearby observed and unobserved nodes to interpolate the value

Table 4: Statistics of all datasets.

Datasets	AQI-36	AQI	NREL-AL	NREL-MD	METR-LA	PEMS-BAY	SEA-LOOP	PEMS07
Nodes	36	437	137	80	207	325	323	883
Time-steps	8,759	8,760	105,120	105,120	34,272	52,128	8,064	28,224
Time interval	1 hour	1 hour	5 min	5 min	5 min	5 min	5 min	5 min
Measurement	PM2.5		Solar power		Traffic Speed			Traffic flow
Data partition	Same as GRIN [5]		Train/Validation/Test: 7/1/2					

**Figure 8: Parameter study on the NREL-MD dataset.**

of each node through an inductive way. It makes two modifications over KCN: (1) It considers temporal information; (2) When constructing the graphs, neighbors of a node are not restricted to observed nodes only.

LSJSTN. LSJSTN [14] further extends IGNNK by improving the aggregation of temporal information through decoupling the model to learn both short-term and long-term temporal patterns.

INCREASE. INCREASE [43] aligns each node with K-nearest observed nodes, designing diverse spatial correlations among nodes, e.g., functionality correlation (spatial similarities based on Point-Of-Interest (POI) data). However, as such POI information is usually unavailable, often INCREASE are solely utilized without POI-information-based graph, as in our experiment.

KITS. KITS [38] is the latest inductive model for spatio-temporal kriging task. By identifying the *graph gap issue* where the graph structure in inference phase has structural difference from the one in training phase, it proposes to utilize virtual nodes to mitigate this influence. However, we empirically show its sensitivity to the missing pattern and assumed missing rate. Besides that, the performance superiority of our model over KITS also show the potential of dynamic and global graphs.

E.3 Evaluations Metrics

We mainly adopt Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE) and Mean Relative Error (MRE) [3] to evaluate the performance of all methods, with the formulas written as follows:

$$\text{MAE} = \frac{1}{N^{\text{unk}}} \sum_i |y_i^{\text{unk}} - \hat{y}_i^{\text{unk}}|$$

$$\text{MAPE} = \frac{1}{N^{\text{unk}}} \sum_i \frac{|y_i^{\text{unk}} - \hat{y}_i^{\text{unk}}|}{|y_i^{\text{unk}}|}$$

$$\text{MRE} = \frac{\sum_i |y_i^{\text{unk}} - \hat{y}_i^{\text{unk}}|}{\sum_i |y_i^{\text{unk}}|}$$

where N^{unk} is the number of unobserved nodes, y^{unk} is the ground truth data, $\hat{y}^{\text{unk}}$ is the estimation generated by kriging models.

E.4 Implementation Details

Following [38], if not specified explicitly, we fix the random seed as 1, and the missing rate $\alpha = 50\%$. For all datasets, the window size in the temporal dimension is 24. The hidden dimension is set as 128. The batch size is set as 32. Besides, we utilize the learning rate scheduler "CosineAnnealingLR". As stated before, the default sampling step is set to $K = 10$, with the validation interval $VI = 5$. The maximum training epoch is 200, with an early stopping mechanism where we stop training after validation performance not been improved for 40 epochs, i.e. $\frac{40}{VI} = 8$ steps of validation. The *transform* can highly influence the model performance. For datasets except METR-LA and PEMS07, $A' = \text{transform}(A)$ should incorporate the global information and fit the form of $QK^T/\sqrt{d}$, therefore *transform* first calculate the length of shortest paths between nodes, as stated before. Then to construct a distance matrix with similar distribution with attention, we assume the distance values of shortest path as the Euclidean distances between high-dimensional vectors, and the weight of A' as the cosine-similarities of the vectors (as the initial states of attention matrices have a scaled variance). $\text{Dis}(i, j) = \sqrt{(\mathbf{v}_i - \mathbf{v}_j)^2, \mathbf{v}_i^2 + \mathbf{v}_j^2 - 2\mathbf{v}_i \cdot \mathbf{v}_j} = \text{Dis}(\mathbf{v}_i, \mathbf{v}_j)^2$. Assuming $\mathbf{v}_i$ is scattered in a ball with $|\mathbf{v}_i| = \text{Mean}(|\mathbf{v}_i|)$, $A'(i, j) = \frac{\mathbf{v}_i \cdot \mathbf{v}_j}{|\mathbf{v}_i||\mathbf{v}_j|}$. For METR-LA and PEMS07, as the connected block in the graph form a abnormal matrix with the shortest path between two nodes far from the usual distances, we simply adopt the matrix from Gaussian Kernel without exceeding outliers. Most experiments are conducted on an NVIDIA GeForce RTX 3080 GPU, with experiments on PEMS07 using NVIDIA GeForce RTX 4090 for its memory cost.

F Parameter Study

We present the parameter studies on NREL-MD datasets as follows in Figure 8. We test how the change of sampling step, number of model layers and number of attention heads influence the performance of the model, showing the stable robustness to parameter choice. The result also verifies our choice of sampling steps as 10 do not significantly influence the performance of reverse process.